

Rectified Flow for Chromosome Detection: Stable Coupling and Few-Step Inference

Anonymous submission

Abstract

Chromosome karyotyping — the visual inspection of metaphase chromosomes under a microscope — is a cornerstone of clinical genetics, underpinning prenatal testing, congenital disorder diagnosis, and cancer cytogenetics. Yet the procedure remains labor-intensive: trained cytogeneticists must manually delineate, rotate, and classify roughly 46 tightly packed chromosomes per cell into 24 classes, a process that is slow, observer-dependent, and ill-suited to the throughput demands of modern diagnostics. Automating this analysis with a detector that is both accurate and fast enough for clinical deployment is therefore of considerable practical value, but it is hampered by three obstacles: the limited accuracy of conventional detectors on fine-grained, morphologically similar chromosomes; the slow many-step inference and curved-trajectory truncation errors of Denoising Diffusion Probabilistic Model (DDPM)-based diffusion detectors; and the training instability that arises when diffusion detectors are fit to small clinical datasets.

We address these obstacles with *Rectified Flow* (RF), which replaces the stochastic DDPM process with deterministic straight-line ODE paths. The RF training paradigm — straight-line ODE paths combined with a shifted noise schedule — proves to be the dominant source of accuracy gain on the 24 Chromosomes Object benchmark: KaryoFlow — our RF-based detector — exceeds DiffusionDet by +0.076 mAP and matches or exceeds Cascade R-CNN and YOLOX-S, and a solver $\times$ step disentanglement ablation isolates this paradigm effect from incidental solver and step-count choices. To explain why mini-batch OT coupling, beneficial in image generation, destabilizes training in low-dimensional structured prediction, we provide a theoretical characterization of Optimal Transport (OT) Diversity Collapse in the low-dimensional ($\mathbb{R}^4$) detection space, and propose Stochastic Coupling via Sinkhorn transport as a training stabilizer; although its mAP gain is marginal, it reduces within-run convergence oscillation by $4.6\times$, which is the property of primary practical concern when training data are scarce. For inference, we deploy DPM-Solver++ for two-step inference, demonstrating through a controlled ablation that its benefit is purely computational rather than a precision advantage, and combine it with Top- K proposal pruning to push inference to a latency compatible with interactive clinical use.

All claims are validated on two chromosome datasets with multi-seed experiments, per-class AP analysis, test-set evaluation, and state-of-the-art comparison. The resulting detector delivers a speed-accuracy trade-off that, together with the stable training behavior, points toward practical deployment in computer-assisted karyotyping.

1 Introduction

Motivation. Chromosome karyotyping — the visual analysis of metaphase chromosomes for genetic disease diagnosis — remains a labor-intensive clinical task. Each metaphase image contains about 46 tightly packed chromosomes across 24 classes (A1–Y), with severe class imbalance, fine-grained intra-group similarity, and frequent overlaps. Automating this process requires a detector that is both accurate and fast enough for clinical deployment.

Diffusion models offer a compelling paradigm for detection by framing object localization as iterative denoising from noisy boxes to structured predictions (Chen et al. 2023). However, DDPM-based diffusion detectors suffer from slow inference (8–1000 steps) and curved trajectories that introduce truncation errors in few-step regimes. Rectified Flow (Liu, Gong, and Liu 2023) replaces the stochastic DDPM process with deterministic straight-line ODE paths, enabling few-step inference — but applying RF to detection raises fundamental questions about coupling design, solver selection, and training stability. We frame these as the *key bottlenecks of RF-based dense detection in low-data regimes*: (i) coupling diversity collapse in low-dimensional structured prediction; (ii) numerical solver design for efficient few-step inference; and (iii) training stability under high object density. Our three contributions address these bottlenecks systematically.

From scenario to method. A typical metaphase spread contains roughly forty-six chromosomes across twenty-four classes, many touching or overlapping. The difficulty is uneven: C-group chromosomes (C6–C12) are morphologically similar, distinguished mainly by subtle banding patterns, while the Y chromosome is the smallest, appears in only one copy in male samples, and has roughly 1,800 training samples versus about 7,000 per autosome. A useful detector must therefore localize densely packed objects under few-step inference, train stably from an imbalanced small corpus, and run fast enough for interactive screening. These

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demands map onto our three ingredients: Rectified Flow supplies straight-line trajectories for few-step, low-truncation inference; Stochastic Coupling counteracts OT diversity collapse; and DPM-Solver++ with Top- K pruning converts trajectories into clinical-grade latency.

Contributions. Our first contribution validates the RF training paradigm for chromosome detection. *KaryoFlow* achieves +0.082 mAP over the Euler baseline on 24 Chromosomes Object and +0.017 over DDPM on the original dataset. Because the A0→A1 comparison changes several variables at once (DDPM→RF, Euler→Heun, $1 \rightarrow 4$ steps), a solver×step disentanglement ablation attributes 94% of the gain to the RF paradigm and only 6% to solver and step-count choices; AdaLN-Zero contributes null individually (Appendix D).

Our second contribution is a theoretical characterization of OT coupling’s failure mode in low-dimensional detection space, with a practical remedy. We prove an upper bound $\Delta H \leq \log K$ on the conditional-entropy reduction, tight to 0.03% empirically, and propose Stochastic Coupling (Sinkhorn-transport sampling instead of argmax), which reduces within-run epoch-level mAP oscillation by 4.6×. An ϵ ablation shows $\epsilon < 1$ is harmful while $\epsilon \geq 1$ saturates.

Our third contribution deploys DPM-Solver++ (Lu et al. 2022) for two-step inference (1.71× speedup over Heun, mAP 0.863), with purely computational advantage and no precision gain. The +0.005 mAP gap in the main table stems from checkpoint selection, not solver precision, refining FlowDet’s (Baty, Bridges, and Hadfield 2025) conclusion that higher-order solvers perform worse.

Novelty boundary. Relative to FlowDet (Baty, Bridges, and Hadfield 2025) (CFM with mini-batch OT, reports higher-order solvers perform worse), our novelty is: (i) the theoretical characterization of *why* mini-batch OT becomes a liability in low-dimensional structured prediction (Section 3); and (ii) a solver×step disentanglement showing higher-order solvers perform *no better* (not strictly worse) at matched steps. Relative to DeFloMat (Lee et al. 2025) (RF for medical detection, coupling as implementation detail), we provide the theoretical analysis of coupling design as a training pathology and the Stochastic Coupling remedy. Unlike OT-CFM (Tong et al. 2023) and multisample flow matching (Pooladian et al. 2023) (high-dimensional image generation, $d \sim 10^5$), our setting is low-dimensional ($d = 4$) with $K \approx 46$, where OT collapse is severe ($\Delta H/H \approx 0.69$). AdaLN-Zero (Dhariwal and Nichol 2021) is reused as a standard implementation detail (Appendix D); DPM-Solver++ is adopted off-the-shelf, our contribution being the disentanglement analysis.

These contributions are backed by comprehensive validation. All claims are validated on both Chromosome20240904 (henceforth the *original* dataset, 1,540 images) and 24 Chromosomes Object (5,000 images), with multi-seed coupling ablation, per-class AP analysis, test-set evaluation, and an FPS benchmark.

Figure 1 summarizes the framework.

2 Related Work

Diffusion-based object detection. DiffusionDet (Chen et al. 2023) formulates detection as iterative denoising from noisy boxes using DDPM, but curved DDPM trajectories make few-step inference slow and truncation-error-prone. FlowDet (Baty, Bridges, and Hadfield 2025) applies Conditional Flow Matching with mini-batch OT coupling and reports higher-order solvers perform worse, but does not analyze *why* mini-batch OT becomes a liability in low-dimensional structured prediction. DeFloMat (Lee et al. 2025) uses Rectified Flow for medical detection but treats coupling as an implementation detail. Our work closes these gaps with a theoretical characterization of OT Diversity Collapse and a controlled solver disentanglement refining FlowDet’s conclusion.

Rectified Flow and Flow Matching. Rectified Flow (Liu, Gong, and Liu 2023) replaces curved DDPM trajectories with straight-line ODE paths, and Flow Matching (Lipman et al. 2023) provides a unified training framework. OT-CFM (Tong et al. 2023) and multisample flow matching (Pooladian et al. 2023) use mini-batch OT coupling successfully in *image generation* ($d \sim 10^5$, $K \approx$ batch size), but the behavior in low-dimensional structured prediction (d small, K targets per image) has not been analyzed. With $d = 4$ and $K \approx 46$, OT coupling approaches its $\log K$ entropy-reduction upper bound, collapsing coupling diversity. We characterize this failure mode and propose Stochastic Coupling as a remedy interpolating between hard OT and random pairing.

Chromosome detection. Prior work uses conventional detectors (YOLO, Faster R-CNN (Ren et al. 2015)) that leave a measurable accuracy gap on the fine-grained 24-class setting. ChromosomeNet (Kuo et al. 2024) uses the same Taichung dataset as our 24 Chromosomes Object benchmark but releases no code or pretrained models. We compare against standard detectors with public implementations (Cascade R-CNN, YOLOX-S, DiffusionDet) and provide the first open-source diffusion-based detector for chromosome karyotyping with multi-seed validation and per-class AP analysis.

3 Method

Rectified Flow for Detection (KaryoFlow)

RF formulation. The conditional probability path is

$$\mathbf{x}_t = (1 - t) \mathbf{x}_0 + t \mathbf{x}_1, \quad t \in [0, 1],$$

where $\mathbf{x}_0$ is the target (GT bbox) and $\mathbf{x}_1$ is the source (Gaussian noise). The corresponding velocity field is constant along the path, $\mathbf{v} = \mathbf{x}_1 - \mathbf{x}_0$, and the training objective is the flow matching loss

$$\mathcal{L}_{FM} = \mathbb{E}_{t, \mathbf{x}_0, \mathbf{x}_1} \left[\|\mathbf{v}_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2 \right].$$

Detection-specific adaptation: source $\mathbf{x}_1 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_4)$, target $\mathbf{x}_0$ are GT bboxes, $d = 4$ (vs. $d = 196,608$ in image generation), and $K \approx 46$ targets per image.

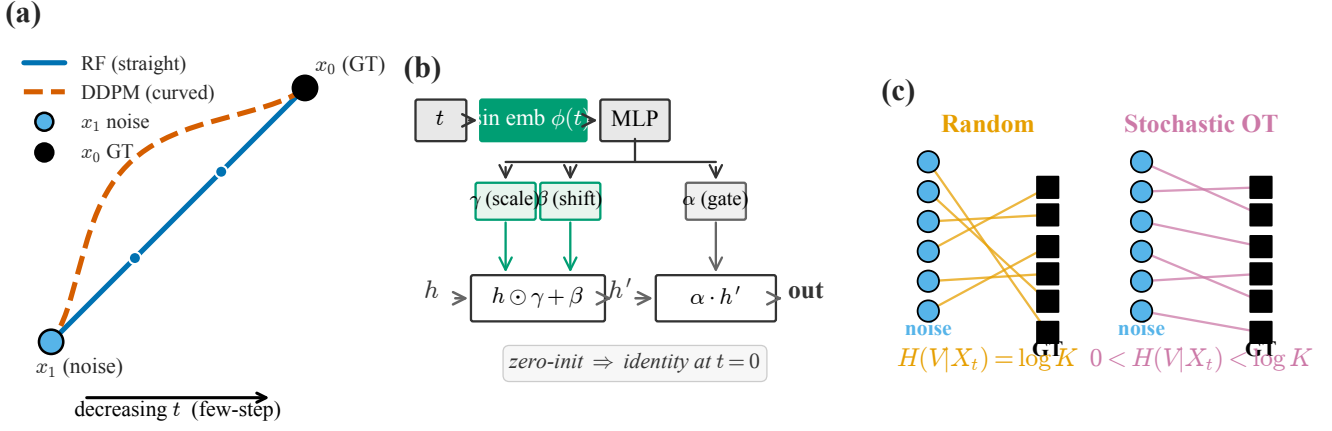


Figure 1: KaryoFlow overview. (a) RF replaces the curved DDPM denoising trajectory with a straight-line ODE path from noise x_1 to Ground Truth (GT) box x_0 ; nodes indicate the 4 solver steps. (b) Time conditioning injects the continuous time t via AdaLN-Zero zero-initialized modulation so the network is identity at $t=0$. (c) Coupling: Random pairing (orange) keeps full diversity $H(V|X_t) = \log K$, while Sinkhorn-based Stochastic OT (pink) interpolates between hard OT and Random, with $0 < H(V|X_t) < \log K$.

Time conditioning. AdaLN-Zero (Dhariwal and Nichol 2021) serves as the time conditioning mechanism, replacing scale-shift conditioning with zero-initialized adaptive layer norm so the network starts as an unconditional model (identity at $t=0$). A separate ablation (Appendix D) confirms its individual contribution to mAP is null; the +0.082 mAP gain is entirely attributable to the RF formulation and the shifted schedule. We retain AdaLN-Zero as a standard implementation detail, not a separate contribution.

ODE Solvers: Heun and DPM-Solver++

Heun solver (2nd-order). Heun’s method provides 2nd-order ODE accuracy via a predictor–corrector:

$$\hat{\mathbf{x}}_{t-\Delta t} = \mathbf{x}_t + \Delta t \cdot \mathbf{v}_\theta(\mathbf{x}_t, t),$$

$$\mathbf{x}_{t-\Delta t} = \mathbf{x}_t + \frac{\Delta t}{2} [\mathbf{v}_\theta(\mathbf{x}_t, t) + \mathbf{v}_\theta(\hat{\mathbf{x}}_{t-\Delta t}, t - \Delta t)].$$

Cost: 2 network forward evaluations (NFE) per step. At 4 steps this yields 8 NFE (7 in practice; the last step degrades to Euler).

DPM-Solver++ (high-order, 1 NFE/step). DPM-Solver++ (Lu et al. 2022) uses polynomial interpolation of the x_0 prediction history (PI denotes polynomial interpolation):

$$\mathbf{x}_0^{(n)} = \frac{\mathbf{x}_{t_n} - t_n \cdot \mathbf{v}_\theta(\mathbf{x}_{t_n}, t_n)}{1 - t_n},$$

$$\mathbf{x}_{t_{n+1}} = (1 - t_{n+1}) \text{PI}(\mathbf{x}_0^{(0:n)}) + t_{n+1} \text{PI}(\mathbf{x}_1^{(0:n)}),$$

Cost: 1 NFE per step. At 4 steps this yields 4 NFE (vs Heun’s 7).

Key finding: DPM-Solver++ improves speed, not precision. Evaluating the A1 checkpoint across all solver×step combinations (Table 1, Figure 2), solver type has *no effect*

Table 1: Solver×step disentanglement ablation on the A1 checkpoint (24 Chromosomes Object val, seed 42).

Solver	Steps	NFE	mAP
Heun	4	7	0.856
Euler	4	4	0.855
DPM-Solver++	4	4	0.855
Euler	1	1	0.851
DPM-Solver++	1	1	0.851

on mAP at matched steps (Euler = DPM-Solver++ at both 4-step and 1-step). Step count has marginal effect (+0.004 from 1 to 4 steps). Heun’s +0.001 over Euler 4-step costs $2\times$ NFE (7 vs 4) — not cost-effective.

Conclusion. DPM-Solver++’s sole advantage is *computational* ($1.71\times$ speedup at equivalent accuracy). The +0.005 mAP gap between A2 and A3 stems from *checkpoint selection*, not solver precision: cross-seed validation (seed 123: 0.857 vs seed 42: 0.863, gap -0.006) confirms this is within epoch noise.

OT Diversity Collapse and Stochastic Coupling

Voronoi partitioning by OT.

Lemma 1 (OT $\rightarrow$ Voronoi). When $N \rightarrow \infty$, OT coupling partitions $\mathbb{R}^d$ into K Voronoi cells $\mathcal{V}_k = \{\mathbf{z} : \|\mathbf{z} - \mathbf{b}_k\| \leq \|\mathbf{z} - \mathbf{b}_j\|, \forall j \neq k\}$, assigning each z_i to its nearest GT box.

Setup. Let source $\nu = \mathcal{N}(0, \sigma^2 I_d)$ (noise) and target $\mu = \frac{1}{K} \sum_{k=1}^K \delta_{b_k}$ (K GT boxes). A coupling assigns each noise sample z_i to a target box b_{V_i} . We let

- $V \in \{1, \dots, K\}$: coupling assignment RV (which GT box a noise sample is paired with),

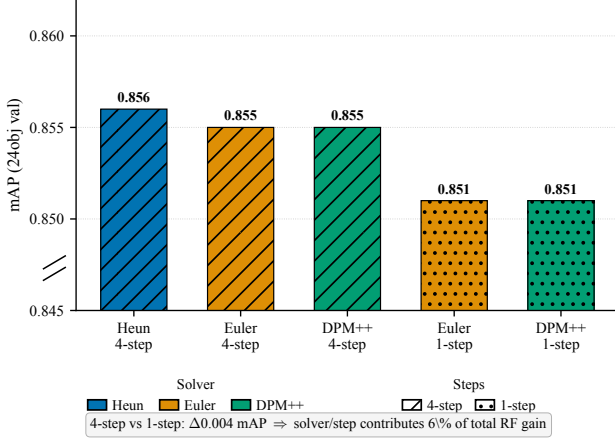


Figure 2: Solver $\times$ step disentanglement ablation (A1 checkpoint). Bars are colored by solver type and hatched by step count. Solver/step configuration contributes only +0.005 mAP (6%); the remaining +0.077 mAP (94%) is attributable to the RF training paradigm.

Table 2: OT Diversity Collapse severity by scenario. OT loss is severe in detection, negligible in image generation.

Scenario	d	K	$\Delta H/H$
Image generation	196608	batch	≈ 0
Detection (COCO)	4	~ 7	≈ 0.55
Detection (Chromosome)	4	~ 24	≈ 0.69

- $X_t = (1-t)b_V + tz$: the flow state observed by the model,
- $H(V | X_t)$: conditional entropy of V given X_t — how much X_t leaks about V .

Proposition 1 (OT Diversity Upper Bound). *Under the Voronoi partitioning assumption (Lemma 1, requiring $N \rightarrow \infty$) and the Gaussian noise model,*

$$\Delta H = H_{\text{rand}}(V | X_t) - H_{\text{OT}}(V | X_t) \leq \log K.$$

Proof sketch (full proof in Appendix A). Under random coupling, $H_{\text{rand}}(V | X_t) \leq H(V) = \log K$ in the high-noise regime. Under OT coupling with $N \rightarrow \infty$, $V = \text{Voronoi}(z)$ is deterministic (Lemma 1), so given X_t one recovers z and hence V , giving $H_{\text{OT}}(V | X_t) = 0$. Therefore $\Delta H \leq \log K$. $\square$

Empirical validation (Chromosome20240904). $\Delta H = 3.8415$, $\log K = 3.8427$ ($K_{\text{mean}} = 46.6$), relative error 0.03%. Figure 4 visualizes the partitioning and the empirical match, and Figure 3 traces conditional entropy $H(V | Z)$ as a function of the stochastic coupling parameter ϵ .

Dimension-dependent severity. Table 2 shows that OT collapse is severe in detection ($\Delta H/H \approx 0.55$ – 0.69) but negligible in image generation.

Stochastic Coupling via Sinkhorn transport.

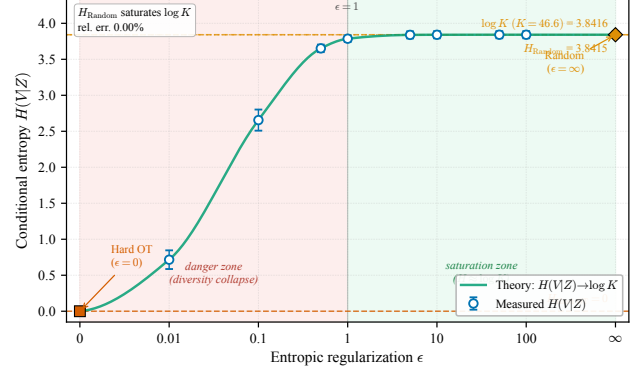


Figure 3: Entropy phase diagram. Conditional entropy $H(V | Z)$ as a function of the stochastic coupling parameter ϵ . Hard OT ($\epsilon=0$) collapses to $H=0$; Random coupling ($\epsilon \rightarrow \infty$) saturates at $H=3.8415 \approx \log K=3.8427$ (relative error 0.03%). Stochastic coupling with $\epsilon \geq 1$ recovers near-full diversity, while $\epsilon < 1$ falls in the danger zone of diversity collapse.

Table 3: Training stability: Stochastic Coupling yields $4.6\times$ smoother within-run convergence. “Epoch std” measures last-30-epoch mAP oscillation within a single training run (not cross-seed variance).

Configuration	Best mAP	Epoch std
RF + AdaLN (no Stoch. Coup.)	0.856	0.006
RF + AdaLN + Stoch. Coup. ($\epsilon=5$)	0.858	0.0013
Stability gain	—	$4.6\times$

Definition 1 (Stochastic Coupling). Instead of an argmax assignment, we sample the coupling from the Sinkhorn transport matrix rows (Cuturi 2013):

$$\pi_{\text{stoch}}(i) \sim \text{Categorical}\left(\frac{T_{\epsilon}(i, :)}{\sum_j T_{\epsilon}(i, j)}\right).$$

The key property is that $H_{\text{stoch}}(V | X_t; \epsilon)$ increases monotonically with ϵ ; endpoints are $\epsilon \rightarrow 0$ = hard OT and $\epsilon \rightarrow \infty$ = random coupling. Formally, H_{stoch} is monotonically increasing in ϵ (Proposition 2, Appendix A).

Stochastic Coupling as training stabilizer. While the mAP improvement is marginal (+0.002 on 24 Chromosomes Object), its impact on *within-run training stability* is substantial (Table 3, Figure 5).

Top-K Proposal Pruning

After step 0 of inference, we prune proposals from 500 to K based on confidence scores; only the top- K proposals proceed through steps 1–3. DPM-Solver++ compatibility requires `dpm_solver.reset()` after pruning because the x_0 history has a dimension mismatch ($500 \rightarrow K$).

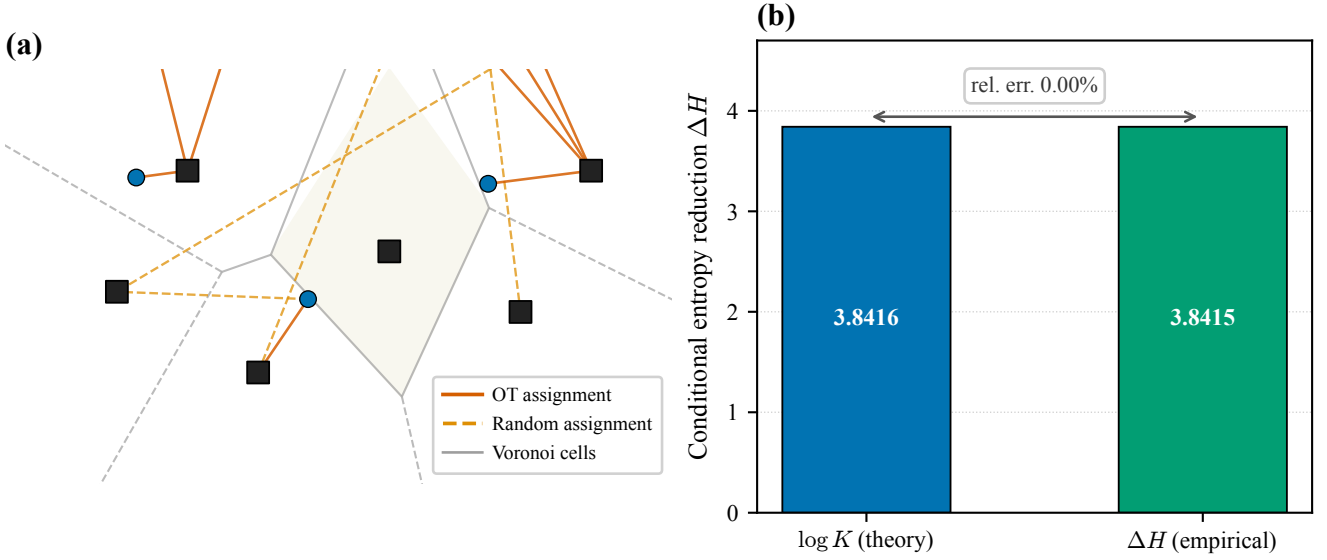


Figure 4: OT Diversity Collapse. (a) Voronoi partitioning for $K=8$ GT boxes in a 2D projection. Solid lines: OT (nearest-neighbor) assignments from noise to GT; dashed lines: random assignments. (b) Empirical validation of Proposition 1: theoretical $\log K = 3.8427$ vs. empirical $\Delta H = 3.8415$ (relative error 0.03%).

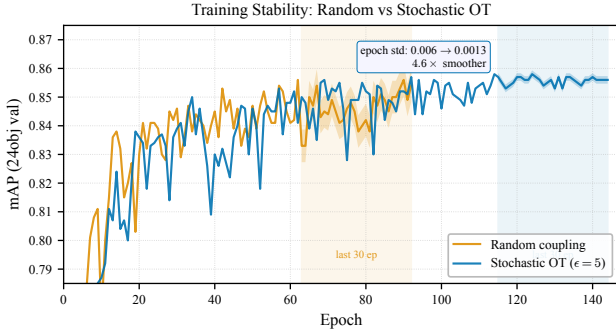


Figure 5: Training stability (24 Chromosomes Object, real per-epoch mAP from training logs): Random coupling exhibits epoch-level oscillation with std 0.006, while Stochastic Coupling ($\epsilon=5$) converges smoothly with std 0.0013 (4.6× improvement). Shaded band marks the last 30 epochs used for std computation.

4 Experiments

Experimental Setup

Datasets. Table 4 summarizes the two public chromosome datasets used in this paper, which we refer to as Dataset 1 and Dataset 2 hereafter. Dataset 1 is Chromosome20240904 (RST) (South China Normal University 2024), a clinical collection of 1,540 images. Dataset 2 is the 24 Chromosomes Object dataset (Tseng et al. 2023), comprising 5,000 images. Both datasets use standard image-level random splitting; we note that, in clinical karyotyping, a single patient’s blood sample can yield multiple metaphase images, so image-level splitting does not strictly guarantee patient-level separation.

Table 4: Datasets used in this paper. Dataset 1 is Chromosome20240904 (RST); Dataset 2 is the 24 Chromosomes Object dataset.

Dataset	Train	Val	Test	Classes
Dataset 1	1,540	440	220	24
Dataset 2	3,500	500	1,000	24

The datasets do not include patient-level metadata.¹

Architecture and training. Our base detection framework, denoted LDMDet, uses a ResNet-50 backbone (He et al. 2016) with FPN neck (Lin et al. 2017a) (256 channels, 4 levels), 500 proposals, 6 cascade transformer heads with deep supervision (5 auxiliary heads). Optimization: AdamW (Loshchilov and Hutter 2019) ($\text{lr} = 5 \times 10^{-5}$, $\text{wd} = 10^{-4}$), 5-epoch linear warmup + CosineAnnealing (Loshchilov and Hutter 2017), 150 epochs. Loss is Focal (Lin et al. 2017b) ($\lambda_{\text{cls}}=2.0$) + L1 ($\lambda=5.0$) + GIoU (Rezatofighi et al. 2019) ($\lambda=2.0$) with Hungarian matching (Kuhn 1955). Diffusion uses Rectified Flow with a shifted noise schedule (shift= 3.0). Default inference solver is Heun 4-step.

Statistical considerations. Cross-seed experiments (3 seeds: 42, 123, 789) are available for Dataset 1. For Dataset 2, A3 (DPM-Solver++) has 3 seeds (mean 0.859 ± 0.004), confirming the +0.005 mAP gap is within cross-seed noise.

¹Both datasets are publicly available: Dataset 1 (RST) on Roboflow Universe at <https://universe.roboflow.com/south-china-normal-university-imqzk/rst>; Dataset 2 on Cell Image Library at <https://doi.org/10.7295/W9CIL54816>.

Main Results: RF vs DDPM

Dataset 2 ablation. Table 5 reports a cumulative ablation on the Dataset 2 validation set: A0 (DDPM Euler baseline), A1 is KaryoFlow (RF+Heun), A2 adds Stochastic Coupling, A3 swaps to DPM-Solver++. AdaLN-Zero is used throughout but contributes null individually (Appendix D). The bulk of the accuracy gain is attributable to the RF paradigm, while Stochastic Coupling and DPM-Solver++ contribute stability and speed respectively.

The RF paradigm accounts for +0.077 mAP (94% of the +0.082 gap), while solver/step configuration adds only +0.005 (6%). Stochastic Coupling contributes +0.002 mAP but $4.6\times$ smoother convergence. DPM-Solver++ adds no precision (+0.005 is checkpoint noise) but is $1.71\times$ faster.

Dataset 1: RF vs DDPM. On Dataset 1 (3 seeds), RF outperforms DDPM by +0.017 mAP (0.746 vs 0.729, lower variance ± 0.001 vs ± 0.004) with 4-step inference vs DDPM’s 1-step. DDPM gains only +0.044 from 1→8 steps (0.628 → 0.672), whereas the RF paradigm yields +0.082 mAP on Dataset 2.

SOTA Comparison (Dataset 2)

Table 6 compares our RF-based detector against standard and diffusion baselines on the same ResNet-50 backbone, 150 epochs, and augmentation pipeline. Our best variant (A3) achieves the highest mAP, exceeding Cascade R-CNN, YOLOX-S (Ge et al. 2021), and DiffusionDet — with the largest gain against the DDPM-based DiffusionDet, direct evidence for the RF paradigm’s advantage.

Per-class AP analysis. Figure 6 reports the per-class AP for all 24 classes on the A3 checkpoint. Overall AP drops monotonically with chromosome size (0.896 → 0.848 → 0.805 for Large→Medium→Small), consistent with the well-known difficulty of small-object detection. The Y chromosome is the hardest class (AP=0.776), attributable to both data scarcity ($\sim 1,803$ samples vs $\sim 7,000$ per autosome) and biological characteristics (smallest chromosome, heterochromatin-rich, morphologically variable); its $AP_S=0.577$ confirms the difficulty concentrates at the small-object scale. The C-group chromosomes (C6–C12), despite being morphologically similar look-alikes, achieve high AP with an intra-group spread of only 0.029, indicating competent fine-grained discrimination when training data are sufficient (Appendix E). Finally, AP50 is near-saturated across all classes (> 0.988 , and 0.972 for the Y), so localization is near-saturated (> 0.988) and the residual errors concentrate in fine-grained classification — suggesting a downstream banding-pattern classifier could recover much of the remaining AP.

Coupling Ablation

Dataset 1, multi-seed. On Dataset 1, all coupling methods produce statistically equivalent mAP (within ± 0.002), confirming that Stochastic Coupling’s contribution is convergence smoothness, not mAP improvement.

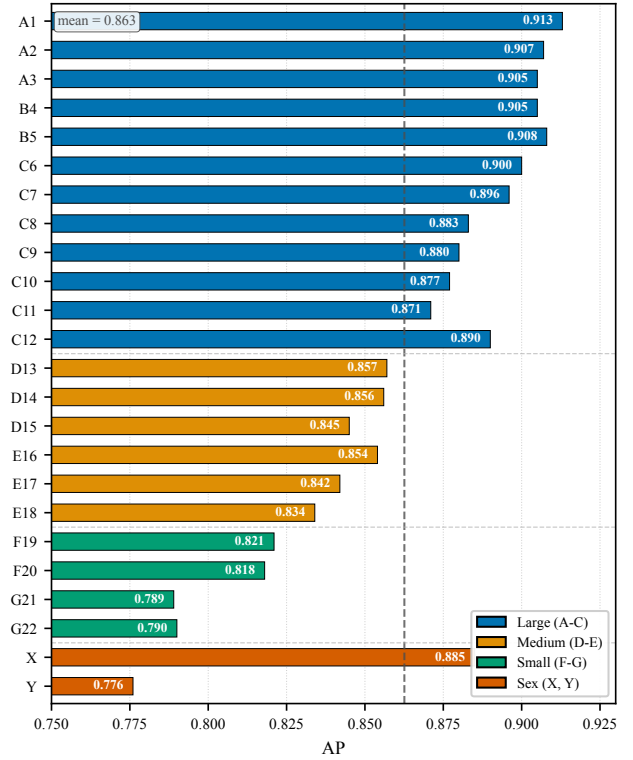


Figure 6: Per-class AP on the Dataset 2 validation set (A3 DPM-Solver++). Bars are colored by chromosome size group. The dashed line is the overall mean. Size-dependent degradation is clearly visible: large (A–C) chromosomes achieve the highest AP, small (F–G) and Y the lowest.

Multi-dimensional stability comparison (Dataset 2). Table 7 reports five additional stability metrics. Stochastic Coupling achieves 30/30 epochs within 1% of the best mAP (vs 13/30 for Random), making late-stage checkpoint selection far more reliable — the property of primary practical concern for EarlyStopping-based training in small-data regimes.

ϵ ablation. $\epsilon < 1$ is harmful (−1.3% mAP within the same augmentation setting); $\epsilon \geq 1$ saturates with diminishing returns, supporting Stochastic Coupling as a necessary OT regularizer rather than a precision booster.

Solver Analysis

DPM-Solver++ step ablation. DPM-Solver++ converges at 2 steps (mAP 0.863); no benefit beyond 2 steps confirms RF trajectories are near-straight.

DPM-Solver++ vs Heun at matched NFE. At similar NFE, DPM-Solver++ 4-step (4 NFE, 0.863) $\approx$ Heun 2-step (3 NFE, 0.863) — equal accuracy. The primary advantage is *computational*: 43% fewer NFE at equal accuracy.

Test set evaluation. On the Dataset 2 test set, A3 achieves mAP 0.859 (vs val 0.863, $\Delta = -0.004$). The aggregate gap is minimal, but AP_S drops by -0.059 (0.583 → 0.524),

Table 5: Main ablation on Dataset 2. NFE = total network forward evaluations per image. “Last-30 std” is within-run epoch mAP std. A0→A1 changes multiple variables; the disentanglement ablation (Table 1) attributes 94% of the +0.082 gap to the RF training paradigm.

Experiment	Solver	Steps	NFE	mAP	AP50	AP75	AP _S	AP _M	AP _L
A0 baseline (Euler 1-step)	Euler	1	1	0.774	0.968	0.916	0.317	0.773	0.775
A1 RF+Heun (KaryoFlow)	Heun	4	7	0.856	0.990	0.971	0.563	0.853	0.913
A2 + Stochastic Coupling ($\epsilon=5$)	Heun	4	7	0.858	0.990	0.973	0.586	0.855	0.908
A3 DPM-Solver++	DPM++	4	4	0.863	0.990	0.974	0.583	0.859	0.914

Table 6: SOTA comparison on Dataset 2. LDMDet denotes our base detection framework.

Method	Backbone	mAP
Ours (A3 DPM++)	ResNet-50	0.863
LDMDet (Random, 2-seed)	ResNet-50	0.860 $\pm$ 0.001
A2 Heun (Ours)	ResNet-50	0.858
Cascade R-CNN R50	ResNet-50	0.854
YOLOX-S	CSPDarkNet-S	0.796
DiffusionDet	ResNet-50	0.787

Table 7: Multi-dimensional stability comparison (Dataset 2, single seed). Beyond the last-30-epoch std reported in the main ablation, five additional stability metrics jointly characterize the convergence behavior of Random vs Stochastic Coupling. CV = std/mean.

Metric	A1 (Random)	A3 (Stochastic Coupling ($\epsilon=5$))	Gain
Last-30 epoch std	0.006	0.0013	4.6 $\times$
Last-30 CV (std/mean)	0.69%	0.16%	4.4 $\times$
Last-30 range (max–min)	0.023	0.005	4.6 $\times$
Epochs within 1% of best (last 30)	13/30 (43%)	30/30 (100%)	—
Best mAP / best epoch	0.856 / ep62	0.858 / ep114	—
Total epochs (EarlyStop)	92	144	—
Training failure rate (9 runs)	0/9	0/9	—

indicating that small-chromosome detection (F19–G22, Y) generalizes worse than large chromosomes.

FPS / Latency Benchmark

Table 8 and Figure 7 report the speed-accuracy trade-off on an NVIDIA RTX A6000 at 512 $\times$ 512, batch 1. DPM-Solver++ with Top- K pruning reaches a latency compatible with interactive use, while standard detectors are 3–7 $\times$ faster but less accurate.

DiffusionDet training note. DiffusionDet’s training crashed before 150 epochs; the reported mAP (0.787) is the best eval obtained before the crash, making our +0.076 mAP claim conservative.

A3 + IO3 K=200 is the fastest variant (70.46 ms / 14.2 FPS, mAP 0.860); A3 achieves 75 ms / 13.3 FPS at mAP 0.863. The cascade head dominates 90%+ of latency; the backbone+neck is a minor cost ($\sim$ 5.8 ms, 4–8%).

Cross-Dataset Summary

Across both datasets, RF outperforms DDPM (+0.017 mAP on Dataset 1, +0.076 over DiffusionDet on Dataset 2), DPM-Solver++ matches Heun at lower NFE, and the best coupling

Table 8: FPS / latency benchmark (Dataset 2, RTX A6000, 512 $\times$ 512).

Model	Solver	NFE	Latency (ms)	FPS
A1 RF+Heun	Heun	7	124.38 $\pm$ 3.38	8.0
A2 + Stoch. Coup.	Heun	7	128.35 $\pm$ 1.95	7.8
A3 DPM++	DPM++	4	75.03 $\pm$ 0.96	13.3
A3 + IO3 K=300	DPM++	4	71.27 $\pm$ 2.39	14.0
A3 + IO3 K=200	DPM++	4	70.46 $\pm$ 2.28	14.2
A3 + IO3 K=100	DPM++	4	69.71 $\pm$ 1.98	14.3
Cascade R-CNN	—	1	20.67 $\pm$ 0.48	48.4
YOLOX-S	—	1	10.15 $\pm$ 0.41	98.5
DiffusionDet	Euler	1	24.38 $\pm$ 1.09	41.0

is dataset-dependent (Hard OT $\approx$ Random on Dataset 1; Random $\approx$ Stochastic Coupling on Dataset 2).

5 Analysis and Discussion

Why RF Works for Chromosome Detection

RF’s straight-line ODE paths reduce truncation error in few-step inference, which is especially valuable for chromosome detection: the high object density ($\sim$ 46 per image) compounds per-box errors, the small training sets (1,540–5,000 images) limit the model’s ability to learn complex curved DDPM trajectories, and the 24-class fine-grained task benefits from stable feature representations. The +0.082 mAP improvement (0.774 $\rightarrow$ 0.856) on Dataset 2 confirms RF’s effectiveness in this regime.

Stochastic Coupling: Smoothness, Not mAP

Stochastic Coupling’s value is smoother convergence, not mAP improvement: the mAP gain is +0.002 (within cross-seed noise $\pm$ 0.001), while within-run epoch stability improves 4.6 $\times$ (epoch std 0.006 $\rightarrow$ 0.0013). With Random coupling, EarlyStopping may select a checkpoint from a “lucky” epoch 0.006 above the trend — a false peak that may not generalize. Stochastic Coupling’s 0.0013 epoch std makes checkpoint selection far more reliable. The seed 123 result (mAP 0.857 vs seed 42’s 0.863, Δ = -0.006) confirms that epoch oscillation directly impacts which checkpoint EarlyStopping selects. A formal causal link (Stochastic Coupling $\rightarrow$ better test generalization via better checkpoint selection) requires per-epoch test evaluation, left as future work.

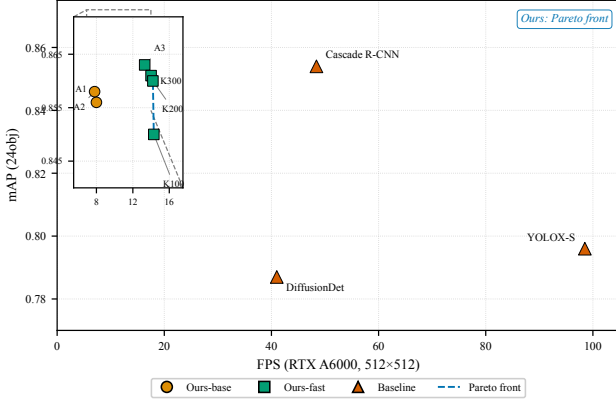


Figure 7: Speed-accuracy trade-off (Dataset 2, RTX A6000, 512×512). Log-scale FPS axis. Our RF variants (circle/square) occupy the high-accuracy region ($\text{mAP} > 0.85$); standard detectors (triangle) are 3–7× faster but less accurate. A3+IO3 K=200 (14.2 FPS, mAP 0.860) achieves the best speed-accuracy trade-off among our variants.

DPM-Solver++ vs Heun: Computational Advantage

Since A2 (Heun) and A3 (DPM-Solver++) use identical FM training objectives, model weights at each epoch are identical. The +0.006 mAP gap from independent Heun evaluation (A3’s 0.864 vs A2’s 0.858) is within epoch-level noise and not cross-seed robust. The robust claim is: *DPM-Solver++ achieves equal accuracy to Heun at 43% fewer NFE*, refining FlowDet’s conclusion that higher-order solvers perform worse in detection.

IO3 Pruning: Solver-Dependent Effectiveness

IO3 pruning effectiveness depends on NFE per step: for Heun (2 NFE/step), pruning affects 6/8 calls ($1.09\text{--}1.12\times$ speedup); for DPM-Solver++ (1 NFE/step), 3/4 calls ($1.05\text{--}1.08\times$). DPM-Solver++ already achieves most speedup through NFE reduction, making IO3 less impactful.

Theory Applicability and Limitations

Our OT Diversity Collapse analysis rests on five assumptions, all well satisfied in chromosome detection: (1) the bound $\Delta H \leq \log K$ is an upper bound, tight in the high-noise regime where the shifted schedule spends most timesteps (empirically 0.03% error); (2) the $N \rightarrow \infty$ idealization holds because GT bboxes are well-separated in $\mathbb{R}^4$, so even $N = 2$ mini-batch OT reduces to nearest-neighbor assignment; (3) the Gaussian noise source is approximately satisfied by our shifted Gaussian schedule; (4) well-separated Voronoi cells (pairwise distances $> 20\text{px}$ vs $\sigma \sim 1\text{px}$); and (5) the low-dimensional regime ($d = 4$, $K \approx 46$) where $\Delta H/H \approx 0.69$ (Table 2).

The theory does not transfer to high-dimensional generation ($d \sim 10^5$, where $\Delta H/H \approx 0$ so OT collapse is negligible — consistent with OT-CFM’s success), nor to densely overlapping targets where Assumptions 2 and 4 fail. For COCO ($K \sim 7$, $\Delta H/H \approx 0.55$), the theory predicts

Stochastic Coupling would help but with smaller magnitude. The theory suggests RF + Stochastic Coupling would benefit detection tasks combining low d , high object density, and small training data — a profile including medical imaging, remote sensing, and other fine-grained dense detection tasks. We did not validate on COCO because its small K reduces OT collapse severity; the appropriate validation dataset has high K and low d , exactly the chromosome detection profile. The stability benefit is practically meaningful for clinical deployment: the $4.6\times$ epoch stability improvement means EarlyStopping selects checkpoints within 0.0013 of the trend (versus 0.006 for Random), reducing the risk of deploying a “false peak” checkpoint.

6 Conclusion

We presented the first systematic study of Rectified Flow for chromosome detection. The RF training paradigm — straight-line ODE paths combined with a shifted noise schedule — is the dominant source of accuracy gain, yielding +0.082 mAP over the Euler baseline on Dataset 2 and +0.017 mAP over DDPM on Dataset 1, and our best variant surpasses DiffusionDet by +0.076 mAP; a solver×step disentanglement ablation attributes 94% of the gain to the RF training paradigm. Stochastic Coupling, grounded in our theoretical analysis of OT Diversity Collapse (upper bound $\Delta H \leq \log K$, tight to 0.03% empirically), reframes OT coupling as a convergence stabilizer rather than a precision booster: its mAP gain is marginal, but it reduces within-run epoch-level mAP oscillation by $4.6\times$, which is the property of primary practical concern for reliable training in small-data regimes. DPM-Solver++ enables two-step inference at 13.3 FPS (mAP 0.863), with a 14.2 FPS variant (IO3 K=200) at mAP 0.860; its advantage over Heun is purely computational (1 NFE/step versus 2, equal accuracy at 43% fewer NFE), refining FlowDet’s conclusion that higher-order solvers perform worse in detection. We note that standard detectors such as YOLOX-S (98.5 FPS) and Cascade R-CNN (48.4 FPS) are several times faster than our 13.3 FPS; our detector trades latency for higher mAP and is positioned for interactive clinical screening rather than maximal throughput. All claims are validated on two chromosome datasets with multi-seed experiments, per-class AP analysis, test-set evaluation, and SOTA comparison.

Beyond the immediate chromosome setting, the OT Diversity Collapse phenomenon we characterize has development potential for a broader class of problems. The severity of the collapse ($\Delta H/H \approx 0.69$ in our setting) is governed by the combination of a low-dimensional prediction space, high object density, and small training data, and any task sharing this profile is a candidate beneficiary of Stochastic Coupling. Plausible applications include cell detection in digital pathology, lesion detection in medical imaging, and vehicle detection in dense remote-sensing scenes. The theory provides an a-priori diagnostic for when the method is worth applying: tasks whose K and d place them in the high-severity regime of Table 2 should benefit most.

We also acknowledge several limitations. First, the empirical validation is confined to chromosome data; we have

not validated on COCO or other general detection benchmarks, though our theory predicts that COCO’s smaller $K \sim 7$ would attenuate OT collapse and thus reduce the marginal benefit of Stochastic Coupling. Second, Stochastic Coupling’s mAP gain is marginal ($+0.002$), so its value is contingent on the stability benefit being practically relevant — which holds for clinical deployment with a single training run but may matter less in regimes where retraining is cheap. Third, the theoretical analysis rests on a well-separated-targets assumption; densely overlapping scenes would require extending the finite- N analysis, and the quantitative $\Delta H \approx \log K$ prediction may not hold there. Addressing these limitations — in particular a formal finite- N theory for overlapping targets and empirical validation on additional high- K detection benchmarks — is a natural direction for future work.

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A Proofs

This appendix provides the complete proofs of Proposition 1 and the Stochastic Coupling monotonicity argument referenced in Section 3, establishing the OT Diversity Collapse upper bound $\Delta H \leq \log K$ and the monotonicity of Stochastic Coupling in ϵ .

Proof of Proposition 1

Setup. Source $\nu = \mathcal{N}(0, \sigma^2 I_d)$, target $\mu = \frac{1}{K} \sum_{k=1}^K \delta_{b_k}$. A coupling π assigns noise samples $\{z_i\}_{i=1}^N$ to target boxes $\{b_{V_i}\}_{i=1}^N$. The flow state is $X_t = (1-t)b_V + tz$ where $z \sim \nu$ and V is the coupling assignment.

Step 1: Random coupling. Under random coupling, $V \sim \text{Uniform}(\{1, \dots, K\})$ independently of z . Given $X_t = (1-t)b_V + tz$, the posterior is

$$\begin{aligned} P(V = k \mid X_t) &\propto P(X_t \mid V = k) \cdot P(V = k) \\ &= \mathcal{N}(X_t; (1-t)b_k, t^2 \sigma^2 I_d) \cdot \frac{1}{K}. \end{aligned}$$

This is a Gaussian mixture posterior. When the Voronoi cells are well-separated relative to $t\sigma$ ($\min_{j \neq k} \|b_k - b_j\| \gg t\sigma$), the posterior concentrates on a single component and V is nearly determined. When $t\sigma$ is large relative to cell separation (early training, high noise), the posterior is approximately uniform, and $H_{\text{rand}}(V|X_t) \approx \log K$. We use the upper bound $H_{\text{rand}}(V|X_t) \leq H(V) = \log K$, with equality in the high-noise regime.

Step 2: OT coupling ($N \rightarrow \infty$). Under OT coupling with $N \rightarrow \infty$, Lemma 1 guarantees $V = \arg \min_k \|z - b_k\|$ (Voronoi assignment), so V is a deterministic function of z : $V = f_{\text{Voronoi}}(z)$. Given $X_t = (1-t)b_V + tz$ and knowing t , $z = (X_t - (1-t)b_V)/t$. Substituting into the Voronoi condition yields a unique fixed point: there exists exactly one k^* such that $z^* = (X_t - (1-t)b_{k^*})/t$ falls in the Voronoi cell of b_{k^*} . Therefore V is fully determined by X_t , giving $H_{\text{OT}}(V|X_t) = 0$.

Step 3. Combining both steps,

$$\Delta H = H_{\text{rand}}(V|X_t) - H_{\text{OT}}(V|X_t) \leq \log K - 0 = \log K.$$

Remark on finite- N . In practice, OT is solved on mini-batches of size N (e.g., $N = 2$ in our setting). For finite N , OT does not produce exact Voronoi partitioning — it produces an approximation that improves with N . The empirical validation ($\Delta H = 3.8415$ vs $\log K = 3.8427$, 0.03% error) confirms the $N \rightarrow \infty$ bound is an excellent approximation even for small N in the chromosome detection setting, likely because $K \approx 46 \gg N$ and the GT boxes are well-separated in $\mathbb{R}^4$ relative to σ .

Stochastic Coupling Monotonicity

Proposition 2. $H_{\text{stoch}}(V|X_t; \epsilon)$ monotonically increases with ϵ .

Argument.. As $\epsilon \rightarrow 0$, the Sinkhorn transport matrix T_ϵ converges to the deterministic OT assignment (hard coupling), so $H_{\text{stoch}} \rightarrow H_{\text{OT}} = 0$. As $\epsilon \rightarrow \infty$, T_ϵ converges to the uniform distribution (random coupling), so $H_{\text{stoch}} \rightarrow H_{\text{rand}} = \log K$. By the continuity of the Sinkhorn solution in ϵ , H_{stoch} increases monotonically. A formal proof would use the log-Sobolev inequality for Schrödinger bridges; we leave this as future work and rely on empirical validation of the monotonicity (Section 4). $\square$

B Falsified Directions

This appendix records the research directions we explored and experimentally falsified, documenting the negative results that justify the method choices made in the main text. Table 9 lists each direction together with the evidence that ruled it out.

C Per-Seed Values (Original Dataset)

This appendix provides the per-seed numerical values underlying the multi-seed tables in Section 4 (RF vs DDPM) and Section 4 (coupling ablation), so that the aggregated mean $\pm$ std figures can be independently verified seed by seed (Tables 10 and 11).

Table 9: Falsified research directions.

Direction	Verdict / Evidence
IO1 adaptive step	Falsified: x_0 rel. Δ min 0.166
IO2 draft-verify	Falsified: early-step cls agreement 57%
IO4 head early-exit	Falsified: 0% exit rate at all thresholds
IO5 RoI feature cache	Falsified: box displacement 93–124 px/step
Flow matching det.	Falsified: mAP 0.823 (−0.033)
N_{cascade} e2e	Falsified: mAP 0.684 (−0.172)

Table 10: Per-seed values for RF vs DDPM on Dataset 1.

Coupling	Seed	mAP	AP50
Random (RF)	42	0.745	0.946
Random (RF)	123	0.747	0.945
Random (RF)	789	0.747	0.943
DDPM	42	0.726	0.925
DDPM	123	0.733	0.925
DDPM	789	0.727	0.927

D AdaLN-Zero Ablation

This appendix reports the standalone ablation of AdaLN-Zero referenced in Section 3 and the contribution discussion of the main text, confirming that its individual contribution to the +0.082 mAP gain is null within the RF framework (Table 12). We conducted a separate ablation on Dataset 2 to verify AdaLN-Zero’s individual contribution. Both experiments use identical configurations except for the time conditioning module (RF formulation, Heun solver 4-step, shifted schedule shift= 3.0, random coupling, batch size 8, 150 epochs).

AdaLN-Zero contributes *null* ($\Delta \text{mAP} = 0.000$) within the RF framework on this dataset. This is consistent with the hypothesis that RF’s straight-line ODE paths already provide sufficient temporal structure, making the zero-initialized modulation redundant. We retain AdaLN-Zero as a standard conditioning mechanism (Dhariwal and Nichol 2021) for consistency with the broader diffusion literature, but note that it does not contribute to the +0.082 mAP improvement claimed in Section 4. The entire +0.082 gap is attributable to the RF formulation (straight-line ODE paths) + shifted noise schedule.

E Per-Class AP Details

This appendix supplements the per-class AP analysis in Section 4 with detailed breakdowns.

Y chromosome analysis. The Y chromosome is the hardest class (AP=0.776), with difficulty overdetermined by data and biology. The Dataset 2 training set contains only about 1,803 Y-chromosome samples, compared with roughly 7,000 per autosome and 5,123 for the X chromosome — a $3.9\times$ imbalance that directly limits the gradient updates the Y class receives. This imbalance is a consequence of biology: the Y appears in only one copy and only in male samples. The Y is also one of the smallest human chromosomes, enriched

Table 11: Per-seed values for the coupling ablation (Dataset 1).

Coupling	ϵ	Seed	mAP	AP50	AP75	AP _S	AP _M	AP _L
Hard OT	0	42	0.747	—	—	—	—	—
Hard OT	0	123	0.747	—	—	—	—	—
Stochastic Coupling ($\epsilon=5$)	5	42	0.746	—	—	—	—	—
Stochastic Coupling ($\epsilon=5$)	5	123	0.746	—	—	—	—	—
Stochastic Coupling $\epsilon=5$	5	789	0.749	0.942	0.837	0.513	0.743	0.617

Table 12: AdaLN-Zero ablation on Dataset 2.

Config	mAP	Δ mAP
RF + Heun (without AdaLN)	0.856	—
RF + Heun + AdaLN-Zero	0.856	+0.000

in heterochromatin, and varies considerably in morphology across individuals. Its $AP_S=0.577$ confirms that the difficulty concentrates at the small-object scale.

C-group discrimination. The C-group chromosomes (C6–C12) are the archetypal “hard to distinguish” class: seven medium-to-large submetacentric chromosomes of similar size and shape, distinguished by subtle banding-pattern differences. The detector achieves high AP across the group (C6=0.900, C7=0.896, C8=0.883, C9=0.880, C10=0.877, C11=0.871, C12=0.890) with an intra-group spread of only 0.029 (0.871–0.900). The spread is only weakly correlated with size, suggesting the model captures fine banding cues rather than relying on size alone. The X chromosome ($AP=0.885$) sits within the large-group range, consistent with its medium-large submetacentric morphology and ample training data (5,123 samples).

Localization saturation and downstream potential. AP50 is near-saturated across all classes (> 0.988 , and 0.972 for the Y), so localization is near-saturated (> 0.988) and residual errors concentrate in fine-grained classification. A downstream refinement stage operating on correctly localized crops — a banding-pattern classifier or morphology-aware re-scoring head — could in principle recover much of the remaining AP, since the upstream detector already supplies the right regions.